Transitivity breaking I The Question

Dataset: $\{((i,i+1),1) \mid i=1,\dots,n-1\} \cup \{((i+1,i),-1) \mid i=1,\dots,n-1\} \cup \{((p,q),1),((q,p),-1)\}$
$p>q$

$$L(w) := \sum_{i=1}^{n-1} (\langle w, g(i,i+1)\rangle - 1)^2 + (\langle w, g(i+1,i)\rangle + 1)^2 + (\langle w, g(p,q)\rangle - 1)^2 + (\langle w, g(q,p)\rangle + 1)^2$$

$$K = \begin{array}{c} (i,i+1) \\ \vdots \\ (i+1,i) \\ \vdots \\ (p,q) \\ (q,p) \end{array} \begin{pmatrix} 1 & 0 & \alpha' & \cdots & \alpha' & (e_p+e_{p-1})\alpha' & (e_q+e_{p-1})\alpha' \\ & \ddots & \ddots & & & & \\ & & 1 & 0 & \alpha' & & \\ & & & 1 & (e_{p-1}+e_q)\alpha' & & (e_{q-1}+e_p)\alpha' \\ & & & & 1 & 0 & \\ & & & & & 1 & \end{pmatrix}, \quad A := \begin{pmatrix} 1 & & & \\ 1 & \ddots & & \\ & \ddots & \ddots & \\ & & 1 & 1 \end{pmatrix}$$

Let $a = \begin{pmatrix} b \\ \tilde{b} \\ c \\ \tilde{c} \end{pmatrix} \Rightarrow b + (A\tilde{b} + c(\delta_p + \delta_{q-1}) + \tilde{c}(\delta_{p-1} + \delta_q))\alpha' = 1$ (1)

$\sim$ w/ $x \leftrightarrow \bar{x}$, $1 \leftrightarrow -1$ ($\bar{1}$)

$(b_p + b_{q-1} + \bar{b}_{p-1} + \bar{b}_q)\alpha' + c = 1$ (2)

$\sim$ w/ $x \leftrightarrow \bar{x}$, $1 \leftrightarrow -1$ ($\bar{2}$)

Solution unique $\Rightarrow b = -\bar{b}$, $c = -\bar{c}$

$\Rightarrow$ (1) $b - Ab\alpha' + c\alpha'(e_p + e_{q-1} - e_{p-1} - e_q) = 1$

(2) $(b_p + b_{q-1} - b_{p-1} - b_q)\alpha' + c = 1$

$\Rightarrow b = (1 - A\alpha')^{-1}(1 - c\alpha'(e_p + e_{q-1} - e_{p-1} - e_q))$ Let $B := (1 - A\alpha')^{-1}$, $\tilde{b} := B1$

$$\boxed{\begin{aligned} \Rightarrow b_i &= \tilde{b}_i - c\alpha'(B_{ip} + B_{iq-1} - B_{i(p-1)} - B_{iq}) \\ c &= 1 - (b_p + b_{q-1} - b_{p-1} - b_q)\alpha' \end{aligned}}$$

II Prerequisites

$$B_{ij} = \frac{\cosh((n-|j-i|)\lambda) - \cosh((n-(i+j))\lambda)}{(1-\alpha)\sinh(\lambda)\sinh(n\lambda)} \qquad (II.1)$$

$$\tilde{b}_j = \frac{\sinh(n\lambda) - 2\sinh(\frac{n}{2}\lambda)\cosh((\frac{n}{2}-j))}{(1-\alpha)\tanh(\lambda/2)\sinh(\lambda)\sinh(\lambda n)} \qquad (II.2)$$

$$\sinh(x) + \sinh(y) = 2\sinh\left(\frac{x+y}{2}\right)\cosh\left(\frac{x-y}{2}\right) \qquad (II.3)$$

$$\sinh(x) - \sinh(y) = 2\cosh\left(\frac{x+y}{2}\right)\sinh\left(\frac{x-y}{2}\right) \qquad (II.4)$$

$$\cosh(x) + \cosh(y) =$$

$$\cosh(x) - \cosh(y) = 2\sinh\left(\frac{x+y}{2}\right)\sinh\left(\frac{x-y}{2}\right) \qquad (II.6)$$

$$f(j,k) = \alpha'(b_j - b_{j-1} + b_{k-1} - b_k) + \alpha' c(\delta_{jp} + \delta_{kq} - \delta_{kp} - \delta_{jq})$$

$$= r_k - r_j,$$

$$r_j := \alpha'(b_{j-1} - b_j + c\,\delta_{jq} - c\,\delta_{jp})$$

$$D_{ij} := \alpha'(B_{(i-1)j} + B_{i(j-1)} - B_{ij} - B_{(i-1)(j-1)})$$

$$\Rightarrow \alpha'(b_{j-1} - b_j) = \underbrace{\alpha'(\tilde{b}_{j-1} - \tilde{b}_j)}_{=\tilde{r}_j} + c(B_{jp} + B_{j(q-1)} - B_{j(p-1)} - B_{jq} - B_{(j-1)p} - B_{(j-1)(q-1)}$$

$$+ B_{(j-1)(p-1)} + B_{(j-1)q}) = \tilde{r}_j + c(D_{jq} - D_{jp})$$

$$\Rightarrow r_j = \tilde{r}_j + c(D_{jq} - D_{jp} + \delta_{jq} - \delta_{jp})$$

$$\tilde{\tilde{r}}_j := \tilde{r}_j + c(D_{jq} - D_{jp})$$

$$c = 1 - (\tilde{\tilde{r}}_q - \tilde{\tilde{r}}_p) = 1 + \tilde{\tilde{r}}_p - \tilde{\tilde{r}}_q = 1 + \tilde{r}_p - \tilde{r}_q + c(D_{qq} - D_{qp} - D_{qp} + D_{pp})$$

$$\Rightarrow c = \frac{1 + \tilde{r}_p - \tilde{r}_q}{1 - D_{qq} - D_{pp} + D_{qp} + D_{pq}}$$

$$\Rightarrow \alpha' B_{ij} = \frac{\alpha' \cosh((n - i\nu_j + i\nu_j)\lambda) - \cosh((n - i\nu_j - i\nu_j)\lambda}{(1-\alpha)\sinh(\lambda)\sinh(n\lambda)} = \frac{\sinh((n - i\nu_j)\lambda)\sinh((i\nu_j)\lambda)}{\sinh(\lambda)\sinh(n\lambda)}$$

$$i \le j \Rightarrow \alpha' B_{ij} = \frac{\sinh((n-j)\lambda)\sinh(i\lambda)}{\sinh(\lambda)\sinh(n\lambda)}$$

Case 1: $i \le j-1$

$$\Rightarrow D_{ij} = \frac{(\sinh((i-1)\lambda) - \sinh(i\lambda))(\sinh((n+1-j)\lambda) - \sinh((n-j)\lambda))}{\sinh(\lambda)\sinh(n\lambda)}$$

$$= \frac{4\cosh((i-\frac{1}{2})\lambda)\sinh(-\frac{1}{2}\lambda)\cosh((n-(j-\frac{1}{2}))\lambda)\sinh(\lambda/2)}{\sinh(\lambda)\sinh(\lambda n)}$$

$$= -2\cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\left(n-\left(j-\tfrac{1}{2}\right)\right)\lambda\right)\frac{\tanh(\lambda/2)}{\sinh(\lambda n)}$$

Case 2: $j \leq i-1$

$$\Rightarrow D_{ij} = -2\cosh\left(\left(j-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\left(n-\left(i-\tfrac{1}{2}\right)\right)\lambda\right)\frac{\tanh(\lambda/2)}{\sinh(\lambda n)}$$

Case 3: $i=j$

$$\Rightarrow D_{ij} = \alpha'\left(2B_{(i-1)i} - B_{(i-1)(i-1)} - B_{ii}\right)$$

$$= \frac{2\sinh((n-i)\lambda)\sinh((i-1)\lambda) - \sinh(i\lambda)\sinh((n-i)\lambda) - \sinh((i-1)\lambda)\sinh((n+1-i)\lambda)}{\sinh(\lambda)\sinh(\lambda n)}$$

$$= \frac{2\sinh((n-i)\lambda)\sinh(-\tfrac{1}{2})\cosh((i-\tfrac{1}{2})\lambda) - \sinh((i-1)\lambda)\sinh(\tfrac{1}{2})\cosh((n+\tfrac{1}{2}-i)\lambda)}{\sinh(\lambda)\sinh(\lambda n)}$$

$$= -\frac{1}{\cosh(\lambda/2)\sinh(\lambda n)}\Big(\sinh\left(\left(n-\left(i-\tfrac{1}{2}\right)\right)\lambda\right)\cosh(\lambda/2)\cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right) -$$

$$\cosh\left(\left(n-\left(i-\tfrac{1}{2}\right)\right)\lambda\right)\sinh(\lambda/2)\cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right) + \sinh\left(\left(i-\tfrac{1}{2}\right)\lambda\right)\cosh(\lambda/2)\cosh\left(\left(n+\tfrac{1}{2}-i\right)\lambda\right)$$

$$- \cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right)\sinh(\lambda/2)\cosh\left(\left(n+\tfrac{1}{2}-i\right)\lambda\right)$$

$$= -\frac{2}{\cosh(\lambda/2)\sinh(\lambda n)}\Big(\cosh(\lambda/2)\sinh(n\lambda) - \sinh(\lambda/2)\cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\left(n+\tfrac{1}{2}-i\right)\lambda\right)$$

$$= 2\frac{\cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\left(n+\tfrac{1}{2}-i\right)\lambda\right)\tanh(\lambda/2)}{\sinh(\lambda n)} - 2$$

$$D_{ij} = \begin{cases} i=j: & -2\left(1 - \dfrac{\cosh\left(\left(i-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\left(n-\left(i-\tfrac{1}{2}\right)\right)\lambda\right)\tanh(\lambda/2)}{\sinh(\lambda n)}\right) \\ i \neq j: & -2\dfrac{\cosh\left(\left(i \vee j-\tfrac{1}{2}\right)\lambda\right)\cosh\left(\left(n-(i \vee j-\tfrac{1}{2})\right)\lambda\right)\tanh(\lambda/2)}{\sinh(\lambda n)} \end{cases}$$

$$d_j := \cosh\left(\left(j-\tfrac{1}{2}\right)\lambda\right), \qquad \bar{d}_j := \cosh\left(\left(n-\left(j-\tfrac{1}{2}\right)\right)\lambda\right)$$

$$\Rightarrow D_{ij} = \begin{cases} i=j: & 2\dfrac{\tanh(\lambda/2)}{\sinh(\lambda n)} d_j \bar{d}_j - 2 \\ i\neq j: & -2\dfrac{\tanh(\lambda/2)}{\sinh(\lambda n)} d_{i\wedge j}\, \bar{d}_{i\vee j} \end{cases}$$

$$e = 1 + 2D_{pq} - D_{qq} - D_{pp} = -2\frac{\tanh(\lambda/2)}{\sinh(\lambda n)}\overbrace{\left(2d_p\bar{d}_q + d_p\bar{d}_p + d_q\bar{d}_q\right)}^{= d_p(\bar{d}_q+\bar{d}_p) + \bar{d}_q(d_p+d_q)} + 5$$

$$d = 1 + \frac{\sinh\left(\left(p-\frac{n+1}{2}\right)\lambda\right) - \sinh\left(\left(q-\frac{n+1}{2}\right)\lambda\right)}{\sinh\left(\left(\frac{n+1}{2}\right)\lambda\right) - \sinh\left(\left(\frac{n-1}{2}\right)\lambda\right)}$$

$d>0$, e depends

d_j monotonic

Case 1: $j<q$

$$r_j = \tilde{r}_j + c(D_{jq} - D_{jp}) = \tilde{r}_j + 2c\frac{\tanh(\lambda/2)}{\sinh(\lambda n)} d_j \overbrace{(\bar{d}_p - \bar{d}_q)}^{<0}$$

Case 2: $j=q$

$$\tilde{r}_j + 2c\frac{\tanh(\lambda/2)}{\sinh(\lambda n)} d_q(\bar{d}_p - \bar{d}_q) + 2c$$

Case 3: $q<j<p$

$$r_j = \tilde{r}_j + 2c\frac{\tanh(\lambda/2)}{\sinh(\lambda n)}(d_q\bar{d}_j - d_p\bar{d}_j)$$

Case 4: $j=p$

$$r_j = \tilde{r}_j + 2c\tanh(\lambda/2)/\sinh(\lambda n)\,\bar{d}_p(d_p - d_q) - 2c$$

Case 5: $j>p$

antitonic

$$r_j = \tilde{r}_j + 2c\tanh(\lambda/2)/\sinh(\lambda n)\,\bar{d}_j\overbrace{(d_p - d_q)}^{>0}$$

$$r_i = \tilde{r}_i - c(D_{ip} - D_{iq})$$

$$c = 1 - r_q + r_p = 1 - \tilde{r}_q + \tilde{r}_p + c(D_{qp} - D_{qq} + D_{pq} - D_{pp}) - 2$$

$$= 1 - \tilde{r}_q + \tilde{r}_p + c(\overbrace{2D_{qp} - D_{qq} - D_{pp}}^{\boxed{1}})$$

$$\boxed{1} = -\frac{2\tanh(\lambda/2)}{\sinh(\lambda n)}\Big(2\cosh((q-\tfrac{1}{2})\lambda)\cosh((n-(p-\tfrac{1}{2}))\lambda) + \cosh((q-\tfrac{1}{2})\lambda)\cosh((n-(q-\tfrac{1}{2}))\lambda)$$

$$+ \cosh((p-\tfrac{1}{2})\lambda)\cosh((n-(p-\tfrac{1}{2}))\lambda)\Big) + 4$$

$$= -\frac{4\tanh(\lambda/2)}{\sinh(\lambda n)}\Big(\cosh((q-\tfrac{1}{2})\lambda)\cosh((n-\tfrac{p+q-1}{2})\lambda)\cosh(\tfrac{p-q}{2}\lambda) +$$

$$\cosh((n-(p-\tfrac{1}{2}))\lambda)\cosh(\tfrac{p+q-1}{2}\lambda)\cosh(\tfrac{p-q}{2}\lambda)\Big) + 4$$

$$4 - 4\cdot\frac{\tanh(\lambda/2)\cosh(\frac{p-q}{2}\lambda)}{\sinh(\lambda n)}\Big(\cosh((q-\tfrac{1}{2})\lambda)\cosh((n-\tfrac{p+q-1}{2})\lambda) + \cosh((n-(p-\tfrac{1}{2}))\lambda)\cosh(\tfrac{p+q-1}{2}\lambda)\Big)$$

$$e := 1 - \boxed{1}$$

$$d := 1 - \frac{\sinh((p-\frac{n+1}{2})\lambda) - \sinh((q-\frac{n+1}{2})\lambda)}{\sinh(\frac{n+1}{2}\lambda) - \sinh(\frac{n-1}{2}\lambda)}$$

$$\Rightarrow c = \frac{d}{e}.$$

$$r_j = \tilde{r}_j + c\,(\underbrace{D_{jq} - D_{jp}}_{= i h_j} + \delta_{jq} - \delta_{jp})$$

Case 1: $j < q$

$$h_j = 2\frac{\tanh(\lambda/2)}{\sinh(\lambda n)}$$

$$D_{ij} = -2\left(1 - \frac{(\cosh(n\lambda) + \cosh((2i-(n+1))\lambda))\sinh(\lambda/2)}{2\sinh(\lambda n)\cosh(\lambda/2)}\right)$$

$$= \quad \frac{\sinh((n-\frac{1}{2})\lambda) + \sinh(\lambda n)\cosh(\lambda/2) - \cosh((2i-(n+1))\lambda)\sinh(\lambda/2)}{\sinh(\lambda n)\cosh(\lambda/2)}$$

$$= \frac{\cosh(\lambda n)\cosh((2i-1)\lambda)\sinh(\lambda/2) - \sinh(\lambda n)\sinh((2i-1)\lambda)\sinh(\lambda/2) - \sinh((n-\frac{1}{2})\lambda)}{\sinh(\lambda n)\cosh(\lambda/2)} -$$

$\sinh(\lambda n)\cosh(\lambda/2)$

$$= -1 - \frac{\sinh(\lambda n)\cosh(\lambda/2) - \cosh(\lambda n)\sinh(\lambda/2) - \cosh((2i-(n+1))\lambda)\sinh(\lambda/2)}{\sinh(\lambda n)\cosh(\lambda/2)}$$

$$=$$

Case 1: $i \neq j$: $\frac{4}{\sinh(\lambda)\sinh(\lambda n)}\left(\cosh((n+1-(i+j))\lambda) + \cosh(|j-i|\lambda)\right)\cosh(\lambda)$

$$D_{ij} = \frac{2}{\sinh(\lambda)\sinh(\lambda n)}\Big(\cosh((n-|j-i-1|)\lambda) + \cosh((n-|j-i+1|)\lambda - 2\cosh((n+1-(i+j))\lambda)$$

$$-2\cosh((n-|j-i|)\lambda) + \cosh((n-|i+j|)\lambda) + \cosh((n+2-(i+j))\lambda)\Big)$$

$$= \frac{4}{\sinh(\lambda)\sinh(\lambda n)}\left(\cosh((n+1-(i+j))\lambda)\cosh(\lambda) + \cosh\left(\frac{k_1\lambda}{2}\right)\cosh\left(\frac{k_2\lambda}{2}\right) - 2\cosh(k_3/2\,\lambda)\cosh(k_4/2\,\lambda)\right.$$

$$R_1 := |j-i-1| + |j-i+1| = |j-(i+1)| + |j-(i-1)| = \begin{cases} j \geq i+1: & 2(j-i) \\ j = i: & 2 \\ j \leq i-1: & 2(i-j) \end{cases}$$

$$-R_2 := |j-(i+1)| - |j-(i-1)| = \begin{cases} j \geq i+1: & -2 \\ j = i: & 0 \\ j \leq i-1: & 2 \end{cases}$$

$$R_3 \quad n+1-(j+i) + (n-|j-i|) = 2n+1-(j+i+|j-i|) = \begin{cases} j \geq i+1: & 2n+1-2j \\ & 2n+1-2i \\ j \leq i-1: & 2n+1-2i \end{cases}$$

$$R_4 \quad n+1-(j+i) - (n-|j-i|) = 1 + |j-i| - (j+i) = \begin{cases} j \geq i+1: & 1-2i \\ & \text{either} \\ j \leq i-1 & 1-2j \end{cases}$$

III Rank expressions

$$f(j,k) = \alpha'(b_j + b_{k-1} - b_{j-1} - b_k) + \alpha' c(\delta_{jp} + \delta_{kq} - \delta_{jq} - \delta_{kp})$$

Test

$$r_j := \tilde{r}_j(b_{j-1} - b_j) + \underbrace{\alpha' c}_{=:\gamma}(\delta_{jq} - \delta_{jp})$$

$$\Rightarrow r_j = \tilde{r}_j - c(\alpha')^2\left(B_{(j-1)p} + B_{(j-1)(q-1)} - B_{jp} - B_{j(q-1)} + B_{j(p-1)} + B_{jq} - B_{(j-1)(p-1)} - B_{(j-1)q}\right)$$

Let $i_{\nu j} = \max(i,j)$, $i_{\lambda j} = \min(i,j)$

$$\Rightarrow B_{ij} = \frac{\cosh((n + i_{\nu j} - i_{\lambda j})\lambda) - \cosh((n - i_{\nu j} - i_{\lambda j})\lambda)}{(1-\alpha)\sinh(\lambda)\sinh(n\lambda)}$$

$$= \frac{\sinh((n - i_{\lambda j})\lambda)\sinh((i_{\nu j})\lambda)}{\alpha'\sinh(\lambda)\sinh(n\lambda)} \qquad \Rightarrow B_{ij} = B_{ji}$$

$$B_{(i-1)j} - B_{ij} + B_{i(j-1)} - B_{(i-1)(j-1)} =: c_{ij}$$

If $i \leq j-1 \Rightarrow$

$$c_{ij} = \frac{2}{\alpha'\sinh(\lambda)\sinh(n\lambda)}\sinh((n-(i-1))\lambda)\cosh((j-\tfrac{1}{2})\lambda)\sinh(\lambda/2) - \sinh((n-i)\lambda)\cosh((j-\tfrac{1}{2})\lambda)\sinh(\lambda/2)$$

$$= \frac{8}{(1-\alpha)\sinh(\lambda)\sinh(n\lambda)}\cosh((n-i+\tfrac{1}{2})\lambda)\cosh((j-\tfrac{1}{2})\lambda)(\sinh(\lambda/2))^2$$

If $i = j \Rightarrow$

$$c_{ii} = \frac{4}{(1-\alpha)\sinh(\lambda)\sinh(n\lambda)}\left(\sinh(i\lambda)\cosh((n+\tfrac{1}{2}-i)\lambda)\sinh(\lambda/2) + \sinh((n+1-i)\lambda)\cosh((i-\tfrac{1}{2})\lambda)\sinh(\lambda/2)\right)$$

$$\Rightarrow r_j = \tilde{r}_j + \underbrace{\gamma(c_{jp} - c_{jq})}_{=:\eta_j}$$

Case 1	Case 2	Case 3	Case 4	Case 5
1	q	p		n

Case 1:

$$\eta_j = c(1-\alpha)\frac{4\sinh(\lambda/2)^2}{\sinh(\lambda)\sinh(n\lambda)}\cosh((n-j+\tfrac{1}{2})\lambda)\sinh(\tfrac{p+q-1}{2}\lambda)\sinh(\tfrac{p-q}{2}\lambda)$$

$$= c\frac{2\alpha}{\sinh(\lambda)\sinh(\lambda n)}\cosh((n-j+\tfrac{1}{2})\lambda)\sinh(\tfrac{p+q-1}{2}\lambda)\sinh(\tfrac{p-q}{2}\lambda)$$

Case 5:

$$\eta_j = c(1-\alpha)\frac{4\sinh(\lambda/2)^2}{\sinh(\lambda)\sinh(n\lambda)}\cosh((j-\tfrac{1}{2})\lambda)\sinh((n-\tfrac{p+q-1}{2})\lambda)\sinh(\tfrac{p-q}{2}\lambda)$$

$$= -c\frac{2\alpha}{\sinh(\lambda)\sinh(\lambda n)}\cosh((j-\tfrac{1}{2})\lambda)\sinh((n-\tfrac{p+q-1}{2})\lambda)\sinh(\tfrac{p-q}{2}\lambda)$$

Case 1

$$\Rightarrow \eta_j = c\frac{4\sinh\left(\frac{p-q}{2}\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}\left(\sinh\left(\left(n+\frac{p+q}{2}-j\right)\lambda\right) + \sinh\left(\left(n+1-\left(j+\frac{p+q}{2}\right)\right)\lambda\right)\right)$$

$$\cosh(\lambda) = \frac{1}{1-q}$$

$$\cosh(\lambda)-1 = \frac{\alpha}{1-q} \Rightarrow \sinh(\lambda/2)^2$$

$$\sinh(\lambda) = \pm\sqrt{\frac{1}{(1-q)^2}-1} = \pm\sqrt{\frac{1-(1-q)^2}{(1-q)^2}}$$

Case 5

$$\Rightarrow \eta_j = c\frac{\alpha\sinh\left(\frac{p-q}{2}\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}\left(\sinh\left(\left(n+j-\frac{p+q}{2}\right)\lambda\right) - \sinh\left(\left(n+1-\left(j+\frac{p+q}{2}\right)\right)\lambda\right)\right)$$

Case 3

$$\Rightarrow c\frac{4\alpha}{\sinh(\lambda)\sinh(\lambda n)}\left(\cosh\left(\left(n-p+\frac{1}{2}\right)\lambda\right)\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right) - \cosh\left(\left(n-j+\frac{1}{2}\right)\lambda\right)\cosh\left(\left(q-\frac{1}{2}\right)\lambda\right)\right)$$

$$= c\frac{4\alpha}{\sinh(\lambda)\sinh(\lambda_n)}\left(\cosh\left(\left(n-p+\frac{1}{2}\right)\lambda\right)\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right) - \cosh(\lambda_n)\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right)\cosh\left(\left(q-\frac{1}{2}\right)\lambda\right) + \sinh(\lambda_n)\sinh\left(\left(j-\frac{1}{2}\right)\lambda\right)\cosh\left(\left(q-\frac{1}{2}\right)\lambda\right)\right)$$

$$= c\frac{4\alpha}{\sinh(\lambda)\sinh(\lambda_n)}\left(\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right)\cosh\left(\left(n-p+\frac{1}{2}\right)\lambda\right) - \cosh\left(\left(j-\frac{1}{2}\right)\lambda\right)\frac{\cosh\left(\left(n+q-\frac{1}{2}\right)\lambda\right)+\cosh\left(\left(n+\frac{1}{2}-q\right)\lambda\right)}{2} + \sinh\left(\left(j-\frac{1}{2}\right)\lambda\right)\frac{\sinh\left(\left(n+q-\frac{1}{2}\right)\lambda\right)+\sinh\left(\left(n+\frac{1}{2}-q\right)\lambda\right)}{2}\right)$$

$$= \frac{2\alpha}{c\sinh(\lambda)\sinh(\lambda_n)}\left(\cosh\left(\left(n+j-p\right)\lambda\right) + \cosh\left(\left(n+1-(j+p)\right)\lambda\right) - \cosh\left(\left(n+q-j\right)\lambda\right)\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right) - \cosh\left(\left(n-(j+q)\right)\lambda\right)\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right)\right)$$

$$= \frac{\alpha}{c\sinh(\lambda)\sinh(\lambda_n)}\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right)\left(\sinh\left(\left(\frac{2n+q-p-j+\frac{1}{2}}{2}\right)\lambda\right)\sinh\left(\frac{j+\frac{1}{2}-(p+q)}{2}\lambda\right)\right)$$

Put a pin in that

IV Figuring out c

$c = 1 - (r_q - r_p)$

Simplify c_{ii}: $c_{ii} = \frac{4\sinh(\lambda/2)}{(1-\alpha)\sinh(\lambda)\sinh(\lambda n)} \underbrace{\left(\sinh(i\lambda)\cosh((n+\tfrac{1}{2}-i)\lambda) + \sinh((n+1-i)\lambda)\cosh((i-\tfrac{1}{2})\lambda)\right)}$

$$= \sinh(i\lambda)\cosh((n+1-i)\lambda)\cosh(\lambda/2) - \sinh(i\lambda)\sinh((n+1-i)\lambda)\sinh(\lambda/2)$$
$$+ \sinh((n+1-i)\lambda)\cosh(i\lambda)\cosh(\lambda/2) - \sinh((n+1-i)\lambda)\sinh(i\lambda)\sinh(\lambda/2)$$
$$= \cosh(\lambda/2)\sinh((n+1)\lambda) - \cosh((n+1)\lambda)\sinh(\lambda/2) + \cosh((n+1-2i)\lambda)\sinh(\lambda/2)$$
$$= \sinh((n+\tfrac{1}{2})\lambda) + \cosh((n+1-2i)\lambda)\sinh(\lambda/2)$$
$$= \sinh(n\lambda)\cosh(\lambda/2) + \sinh(\lambda/2)\cosh(n\lambda) + \sinh(\lambda/2)\cosh((n+1-2i)\lambda)$$

$$\Rightarrow c_{ii} = \frac{2\cosh(\lambda)}{(1-\alpha)\sinh(\lambda)} + \frac{2\alpha}{(1-\alpha)^2\sinh(\lambda)\sinh(\lambda n)}\left(\cosh(n\lambda) + \cosh((n+1-2i)\lambda)\right)$$

Alt version of c_{ij}

$$\sinh(\lambda) = \sqrt{\frac{1}{(1-\alpha)^2} - 1} = \frac{\sqrt{1-(1-\alpha)^2}}{(1-\alpha)} = \frac{\sqrt{2\alpha - \alpha^2}}{1-\alpha} = \frac{\alpha}{1-\alpha}\sqrt{\frac{2}{\alpha} - 1}$$

$$\Rightarrow c_{ij} = \frac{2\alpha}{(1-\alpha)^2\sinh(\lambda)\sinh(n\lambda)}\left(\cosh((n-(i-j))\lambda) + \cosh((n+1-(i+j))\lambda)\right)$$

$$\Rightarrow c_{ij} = \frac{2\alpha}{(1-\alpha)^2\sinh(\lambda)\sinh(\lambda n)}\left(\cosh((n+|j-i|)\lambda) + \cosh((n+1-(i+j))\lambda)\right)$$

$$\frac{1}{1-\alpha} = \cosh(\lambda) \Rightarrow \frac{1}{1-\cosh(\lambda)}$$

$$\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right) =$$
$$= \cosh(\lambda n)\sinh(\lambda)$$

$$\boxed{\begin{aligned} \tilde{c}_{ij} &= \frac{2\alpha}{(1-\alpha)^2\sinh(\lambda)\sinh(\lambda n)}\left(\cosh((n+|j-i|)\lambda) + \cosh((n+1-(i+j))\lambda)\right) \\ c_{ij} &= \begin{cases} \tilde{c}_{ij} & \text{if } i \neq j \\ \tilde{c}_{ii} + \frac{2}{(1-\alpha)^2\sinh(\lambda)} & \text{if } i = j \end{cases} \end{aligned}}$$

$$r_j = \tilde{r}_j + \frac{(1-\alpha)^2}{4}(c_{jp} - c_{jq})$$

$$\tilde{\tilde{r}} = \tilde{r}_j + \frac{c\alpha}{2}\,\frac{\cosh((n+|j-p|)\lambda) - \cosh((n+|j-q|)\lambda) + \cosh((n+1-(j-p))\lambda) - \cosh((n+1-(j+q))\lambda)}{\sinh(\lambda)\sinh(\lambda n)}$$

Attempt 1

$$= \tilde{r}_j + \frac{c\alpha}{2}\,\frac{\sinh\left(\left(n+\frac{|j-p|+|j-q|}{2}\right)\lambda\right)\sinh\left(\frac{|j-p|-|j-q|}{2}\lambda\right) + \sinh\left(\left(n+1-j-\frac{p+q}{2}\right)\lambda\right)\sinh\left(\frac{q-p}{2}\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}$$

Case 1a: $j \le q$

$$= \tilde{r}_j + \frac{c\alpha}{2}\sinh\left(\left(n+\frac{p+q}{2}-j\right)\lambda\right)\sinh\left(\frac{p-q}{2}\lambda\right) + \dots$$

$$= \tilde{r}_j + c\alpha\,\frac{\sinh\left(\frac{p-q}{2}\lambda\right)\cosh\left(\left(n+\frac{1}{2}-j\right)\lambda\right) - \sinh\left(\left(\frac{p+q}{2}-\frac{1}{2}\right)\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}$$

$\sinh(a-b)\cosh(c) - \sinh(a-c)\cosh(b)$

$\cosh(a-b)\sinh(c+b) - \cosh(a)\sinh(c)$

Case 1b: $j \ge p$

$$= \tilde{r}_j + \frac{c\alpha}{2}\sinh\left(\left(n+j-\frac{p+q}{2}\right)\lambda\right)\sinh\left(\frac{q-p}{2}\lambda\right) \dots =$$

$$= \tilde{r}_j - c\alpha\,\frac{\sinh\left(\frac{p-q}{2}\lambda\right)\sinh\left(\left(n+\frac{1}{2}-\frac{p+q}{2}\right)\lambda\right)\cosh\left(\left(j-\frac{1}{2}\right)\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}$$

Case 1c: $q \le j \le p$

$$= \tilde{r}_j + \frac{c\alpha}{2}\sinh\left(\frac{p-q}{2}\lambda\right)\sinh\left(\left(\frac{p+q}{2}-j\right)\lambda\right) + \dots$$

$$= \tilde{\tilde{r}}_j + c\alpha\,\frac{\sinh\left(\frac{p-q}{2}\lambda\right)\cosh\left(\left(j-\frac{n+1}{2}\right)\lambda\right)\sinh\left(\left(\frac{p+q}{2}-\frac{n+1}{2}\right)\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}$$

$$r_p - r_q = \tilde{\tilde{r}}_p - \tilde{\tilde{r}}_q - c$$

$$\tilde{\tilde{r}}_p - \tilde{\tilde{r}}_q = \frac{\sinh\left(\left(p - \frac{n+1}{2}\right)\lambda\right) - \sinh\left(\left(q - \frac{n+1}{2}\right)\lambda\right)}{\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right)} +$$

$$\alpha \frac{\sinh\left(\frac{p-q}{2}\lambda\right)\sinh\left(\left(\frac{p+q}{2} - \frac{n+1}{2}\right)\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}\left(\cosh\left(\left(q - \frac{n+1}{2}\right)\lambda\right) - \cosh\left(\left(p - \frac{n+1}{2}\right)\lambda\right)\right)$$

$$= \frac{\cosh\left(\left(\frac{p+q}{2} - \frac{n+1}{2}\right)\lambda\right)\sinh\left(\frac{p-q}{2}\lambda\right)}{\cosh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right)}$$

$$+2c\alpha \frac{\sinh^2\left(\frac{p-q}{2}\lambda\right)\sinh^2\left(\left(\frac{p+q}{2} - \frac{n+1}{2}\right)\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}$$

$$c = 1 + r_p - r_q = \tilde{\tilde{r}}_p - \tilde{\tilde{r}}_q + 1$$

$$\Leftrightarrow \left(1 - 2\alpha \frac{\sinh^2\left(\frac{p-q}{2}\lambda\right)\sinh^2\left(\left(\frac{p+q}{2} - \frac{n+1}{2}\right)\lambda\right)}{\sinh(\lambda)\sinh(\lambda n)}\right) c = \frac{\cosh\left(\left(\frac{p+1}{2} - \frac{n+1}{2}\right)\lambda\right)\sinh\left(\frac{p-q}{2}\lambda\right)}{\cosh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right)} + 1$$

Attempt 1a

$\Rightarrow$ Case 1a: $\cosh((n-(j-\frac{1}{2}))\lambda) = \cosh(\lambda n)\cosh((j-\frac{1}{2})\lambda) - \sinh(\lambda n)\sinh((j-\frac{1}{2})\lambda)$

$\Rightarrow \sinh((\frac{p+q}{2}-\frac{1}{2})\lambda)\cosh((n-(j-\frac{1}{2}))\lambda) = \frac{1}{2}\Big(\sinh((n+\frac{p+q}{2}-\frac{1}{2})\lambda)\cosh((j-\frac{1}{2})\lambda) +$

$\sinh((\frac{p+q}{2}-\frac{1}{2}-n)\lambda)\cosh((j-\frac{1}{2})\lambda) - \cosh((n+\frac{p+q}{2}-\frac{1}{2})\lambda)\sinh((j-\frac{1}{2})\lambda) +$
$\cosh((\frac{p+q}{2}-\frac{1}{2}-n)\lambda)\sinh((j-\frac{1}{2})\lambda)\Big)$

$= \frac{1}{2}\Big(\sinh((n+\frac{p+q}{2}-j)\lambda) - \sinh(($

$\sinh(a-b)\cosh(c) - \sinh(a-c)\cosh(b)$
$= \sinh(a)\cosh(b)\cosh(c) - \cosh(a)\sinh(b)\cosh(c)$
$- \sinh(a)\cosh(b)\cosh(c) + \cosh(a)\sinh(c)\cosh(b)$
$= \cosh(a)\sinh(c-b)$

$\cosh(a-b)\sinh(c+b)$ $= \cosh(a)\cosh(b)\sinh(c+b) - \sinh(a)\sinh(b)\sinh(b+c)$

$= \cosh(a)\cosh^2(b)\sinh(c) + \cosh(a)\sinh(b)\cosh(b)\sinh(c)$
$- \sinh(a)\sinh^2(b)\cosh(c) - \sinh(a)\sinh(b)\cosh(b)\sinh(c) - \cosh(a)\sinh(c)$

$= \cosh(a)(\cosh^2(b) - 1)\sinh(c) + \ldots$
$+ (\cosh(a)\sinh(c) - \sinh(a)\cosh(c))(\sinh^2(b) + \sinh(b)\cosh(b))$
$= \sinh(c-a)(\cosh(2b)/2 - \frac{1}{2} + \sinh(2b)/2) = \sinh(c-a)(e^{2b} - \frac{1}{2})$

$$r_i = \tilde{r}_i + \frac{c(1-\alpha)^2}{4}\left(c_{ip} - c_{iq}\right)$$

$$r_q - r_p = \tilde{r}_q - \tilde{r}_p + \frac{c(1-\alpha)^2}{4}\left(c_{qp} - c_{qq} - c_{pp} + c_{pq}\right)$$

$$= \tilde{r}_q - \tilde{r}_p + \frac{c\alpha}{2\sinh(\lambda)\sinh(\lambda n)}\Big(2\big(\cosh((n+(p-q))\lambda) + \cosh((n+1-(p+q))\lambda)\big)$$
$$\underbrace{- 2\cosh(\lambda n) - \big(\cosh((n+1-2p)\lambda) + \cosh((n+1-2q))\big)}$$

$$= \Big[4\cosh\big((n+\tfrac{1}{2}-q)\lambda\big)\cosh\big((\tfrac{1}{2}+p)\lambda\big) - 2\cosh(\lambda n)$$
$$- 2\cosh(n+1-(p+q))\cosh((p-q)\lambda)$$

$$2\big(\cosh(n\lambda) - \cosh((n+1-(p+q))\lambda)\big)\cosh((p-q)\lambda)$$
$$+ 2\sinh(n\lambda)\sinh((p-q)\lambda) - 2\big(\cosh(n\lambda) - \cosh((n+1-(p+q))\lambda)\big)$$
$$= 2\big(\cosh(n\lambda) - \cosh((n+1-(p+q))\lambda)\big)\big(\cosh((p-q)\lambda) - 1\big)$$
$$+ 2\sinh(n\lambda)\sinh(\lambda)$$
$$= 4\sinh\Big(\big(n+\tfrac{1}{2}-\tfrac{p+q}{2}\big)\lambda\Big)\sinh\Big(\underbrace{\big(\tfrac{p+q}{2}-\tfrac{1}{2}\big)}_{\#_1}\lambda\Big)\big(\cosh((p-q)\lambda)\quad\big)$$
$$+ 2\sinh(n\lambda)\sinh(\lambda)$$
$$-1$$

$$\Rightarrow c = 1 - (r_q - r_p) = 1 - \Bigg(\frac{\sinh\big((q-\frac{n+1}{2})\lambda\big) - \sinh\big((p-\frac{n+1}{2})\lambda\big)}{\sinh(\frac{n+1}{2}\lambda) - \sinh(\frac{n-1}{2}\lambda)} +$$
$$\frac{c\alpha}{\sinh(\lambda)\sinh(\lambda n)}\big(\cosh(n\lambda) - \cosh((n+1-(p+q))\lambda)\big)\big(\cosh((p-q)\lambda) - 1\big)$$
$$+ c\alpha\Bigg)$$

$$\#_1 = 2\,\frac{\cosh\big(\frac{n+1}{2} - (\frac{p+q}{2})\big)\sinh(q-p)}{\cosh(\frac{n\lambda}{2})\cosh(\lambda/2)} = 2\,\frac{\big(\cosh(\frac{n+1}{2})\cosh(\frac{p+q}{2}) - \sinh(\frac{n+1}{2})\sinh(\frac{p+q}{2})\big)\sinh(q-p)}{\cosh(n\lambda/2)\cosh(\frac{\lambda}{2})}$$

$$= 2\big(\cosh(\tfrac{p+q}{2})(1 + \tanh(n/2\lambda)\tanh(\lambda/2)\big)\sinh(p-q) - \big(\tanh(\tfrac{n}{2}) + \tanh(\tfrac{\lambda}{2})\big)\sinh(\tfrac{p+q}{2})\sinh(q-p)$$

$$\#_2 = \frac{4c\alpha\sinh\big(n+\frac{1}{2}-\frac{p+q}{2}\big)\sinh\big(\frac{1}{2}-\frac{p+q}{2}\big)\sinh^2((p-q)\lambda)}{\sinh(\lambda)\sinh(\lambda n)}$$

We take brute force solution for c for now!

$$c = 1-(r_q - r_p) = 1 - \left(\frac{\sinh\left(\left(q-\frac{n+1}{2}\right)\lambda\right) - \sinh\left(\left(p-\frac{n+1}{2}\right)\lambda\right)}{\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right)} + \frac{c\alpha}{\sinh(\lambda)\sinh(\lambda n)}\left(\cosh(n\lambda) - \cosh\left(\left(n+1-(p+q)\right)\lambda\right)\right)\left(\cosh\left((p-q)\lambda\right)-1\right) + c\alpha \right)$$

$$\Rightarrow c = \frac{\left(\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right) - \sinh\left(\left(q-\frac{n+1}{2}\right)\lambda\right) + \sinh\left(\left(p-\frac{n+1}{2}\right)\lambda\right)\right)\sinh(\lambda)\sinh(\lambda n)}{\alpha\left(\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh\left(\frac{n-1}{2}\lambda\right)\right)\left(\left(\cosh(n\lambda) - \cosh\left(\left(n+1-(p+q)\right)\lambda\right)\right)\left(\cosh\left((p-q)\lambda\right)-1\right)+1\right)}$$

$$= \frac{\left(2\cosh\left(\frac{n}{2}\lambda\right)\sinh(\lambda/2) - 2\cosh\left(\left(\frac{n+1}{2} - \frac{(p+q)}{2}\right)\lambda\right)\sinh\left(\frac{q-p}{2}\lambda\right)\right)\sinh(\lambda)\sinh(\lambda n)}{2\alpha\cosh\left(\frac{n}{2}\lambda\right)\sinh(\lambda/2)\left(2\sinh\left(\left(n+\frac{1}{2}-\frac{p+q}{2}\right)\lambda\right)\sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\frac{p-q}{2}\lambda\right)+1\right)}$$

$$= \frac{A\sinh(\lambda)\sinh(\lambda n)}{2\alpha\cosh\left(\frac{n}{2}\lambda\right)\sinh(\lambda/2)B}$$

$$A = 2\cosh\left(\frac{n}{2}\lambda\right)\sinh(\lambda/2) - 2\cosh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\frac{1-p}{2}\lambda\right) + 2\sinh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{p+q-1}{2}\lambda\right)\sinh\left(\frac{q-p}{2}\lambda\right)$$

$$= \cosh\left(\frac{n}{2}\lambda\right)\left(2\sinh(\lambda/2) - \sinh\left(\left(q-\frac{1}{2}\right)\lambda\right) - \sinh\left(\left(p-\frac{1}{2}\right)\lambda\right)\right) + \sinh\left(\frac{n}{2}\lambda\right)\left(\cosh\left(\left(q-\frac{1}{2}\right)\lambda\right) - \cosh\left(\left(p-\frac{1}{2}\right)\lambda\right)\right)$$

$$= \cosh\left(\frac{n}{2}\lambda\right)\left(2\sinh(\lambda/2) - \cosh(\lambda/2)\left(\sinh(q\lambda)+\sinh(p\lambda)\right) + \sinh(\lambda/2)\left(\cosh(p\lambda)+\cosh(q\lambda)\right)\right) + \sinh\left(\frac{n}{2}\lambda\right)\left(\cosh(\lambda/2)\left(\cosh(q\lambda)-\cosh(p\lambda)\right) - \sinh(\lambda/2)\left(\sinh(q\lambda)-\sinh(p\lambda)\right)\right)$$

$$= \cosh(p\lambda)\left(\cosh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right) - \sinh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right)\right) + \cosh(q\lambda)\left(\cosh\left(\frac{\lambda}{2}\right)\sinh\left(\frac{n}{2}\lambda\right) + \sinh(\lambda/2)\cosh\left(\frac{n}{2}\lambda\right)\right)$$
$$+ \sinh(p\lambda)\left(\sinh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right) - \cosh\left(\frac{\lambda}{2}\right)\cosh\left(\frac{n}{2}\lambda\right)\right) - \sinh(q\lambda)\left(\sinh\left(\frac{n}{2}\lambda\right)\sinh\left(\frac{\lambda}{2}\right) + \cosh\left(\frac{n}{2}\lambda\right)\cosh\left(\frac{\lambda}{2}\right)\right)$$
$$+ 2\sinh\left(\frac{\lambda}{2}\right)\cosh\left(\frac{n}{2}\lambda\right)$$

$$= 2\sinh\left(\frac{\lambda}{2}\right)\cosh\left(\frac{n}{2}\lambda\right) - \cosh(p\lambda)\sinh\left(\frac{n-1}{2}\lambda\right) + \cosh(q\lambda)\sinh\left(\frac{n+1}{2}\lambda\right) - \sinh(p\lambda)\cosh\left(\frac{n-1}{2}\lambda\right) - \sinh(q\lambda)\cosh\left(\frac{n+1}{2}\lambda\right)$$

$1 \cdot \cosh(\lambda)$

$$\tilde{m} = \sinh(\lambda)\sinh(\lambda n) - \overbrace{\cosh(\lambda)\cosh(n\lambda)\sinh^2\left(\frac{p-q}{2}\lambda\right)}^{(**)}$$
$$+ 2\cosh((n+1-(p+q))\lambda)\cosh(\lambda)\sinh^2\left(\frac{p-q}{2}\lambda\right)$$

$$\tilde{\sim} = 4\cosh(\lambda/2)\cosh(\lambda)\sinh(n/2\,\lambda)\cosh\left(\frac{n+1-(p+q)}{2}\lambda\right)\sinh\left(\frac{q-p}{2}\lambda\right)$$

$$(**) = \cosh(\lambda)\sinh\left(\frac{p-q}{2}\lambda\right)\left(\sinh\left(\left(n+\frac{p-q}{2}\right)\lambda\right) + \sinh\left(\left(n-\frac{p-q}{2}\right)\lambda\right)\right)$$

$=$

$$\sinh^2(x) = \frac{\cosh(2x)-1}{2}$$
$$\sinh(2x) = 2\sinh(x)\cosh(x)$$

Let's look at $\tilde{C}^{-1} := (1-\alpha)\frac{4\sinh(\lambda/2)^2}{\sinh(\lambda)\sinh(n\lambda)} = \frac{4\sinh(\lambda/2)^2}{\sinh(\lambda)\cosh(\lambda)\sinh(n\lambda)}$

$$\Rightarrow \tilde{C} = \frac{\sinh(\lambda)\cosh(\lambda)\sinh(n\lambda)}{4\sinh(\lambda/2)^2} = C\frac{\cosh(\lambda)\cosh(\lambda/2)\sinh(\frac{n\lambda}{2})}{\sinh(\lambda/2)^2} =: C\tilde{\tilde{C}}$$

Case 3: $\eta_j = \frac{2c}{\tilde{C}}\left(\cosh((n-(j-\frac{1}{2}))\lambda)\cosh((p-\frac{1}{2})\lambda) - \cosh((n-q+\frac{1}{2})\lambda)\cosh((j-\frac{1}{2})\lambda)\right)$

Case 2: $\eta_q = \frac{2c}{\tilde{C}}\left(\cosh((n-(j-\frac{1}{2}))\lambda)\cosh((p-\frac{1}{2})\lambda)\right) -$

Let's give c_{ii} another shot:

$$c_{ii} = \frac{2}{\tilde{C}\sinh(\lambda/2)}\left(\sinh(i\lambda)\cosh((n+\tfrac{1}{2}-i)\lambda) + \sinh((n+1-i)\lambda)\cosh((i-\tfrac{1}{2})\lambda)\right)$$

$$= \frac{2}{\tilde{C}\sinh(\lambda/2)}\Big(\cosh((n+1-i)\lambda)\sinh(i\lambda)\cosh(\lambda/2) - \sinh((n+1-i)\lambda)\sinh(i\lambda)\sinh(\lambda/2) + \cosh(i\lambda)\sinh((n+1-i)\lambda)\cosh(\lambda/2) - \sinh(i\lambda)\sinh((n+1-i)\lambda)\sinh(\lambda/2)\Big)$$

$$= \frac{2}{\tilde{C}\sinh(\lambda/2)}\left(\cosh(\lambda/2)\sinh((n+1)\lambda) - 2\sinh(\lambda/2)\sinh(i\lambda)\sinh((n+1-i)\lambda)\right)$$

$$= \frac{2}{\tilde{C}\sinh(\lambda/2)}\left(\sinh((n+1)\lambda)(\cosh(\lambda/2) - \sinh(\lambda/2)) - \sinh((n+1-2i)\lambda)\sinh(\lambda/2)\right)$$

Redo c_{ij}?

$$\Rightarrow c_{ij} = \frac{2}{\tilde{C}}\cosh((n-i+\tfrac{1}{2})\lambda)\cosh((j-\tfrac{1}{2})\lambda)$$

$$= \frac{2}{\tilde{C}}\Big(\cosh((n-i+1)\lambda)\cosh(\lambda/2) - \sinh((n-i+1)\lambda)\sinh(\lambda/2)\Big)\Big(\cosh(j\lambda)\cosh(\lambda/2) - \sinh(j\lambda)\sinh(\lambda/2)\Big)$$

$$= \frac{2}{\tilde{C}}\Big(\cosh((n-i+1)\lambda)\cosh(j\lambda) + \sinh^2(\lambda/2)\cosh((n+1-(i+j))\lambda) + \sinh(\lambda/2)\cosh(\lambda/2)\sinh((i+j-(n+1))\lambda)\Big)$$

$$= \frac{2}{\tilde{C}}\Big(\cosh((n+1-(i+j))\lambda)(\tfrac{1}{2}+\sinh^2(\lambda/2)) + \tfrac{1}{2}\cosh((n+1-(i-j))\lambda) + \frac{\sinh(\lambda)}{2}\sinh((i+j)-(n+1))\lambda)\Big)$$

$$= \frac{1}{c}\left(\cosh(\lambda)\cosh((n+1-(i+j))\lambda) - \sinh(\lambda)\sinh((n+1-(i+j))\lambda) + \cosh((n+1-(i-j))\lambda)\right.$$

$$= \frac{1}{c}\left(\sinh((n-(i+j))\lambda) + \cosh((n+1-(i-j))\lambda)\right.$$